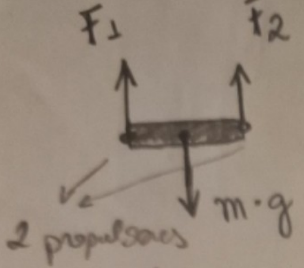
Drone 2D - Modelagem



massa m

mom. de Inércia I

$\frac{1}{12} m L^2$

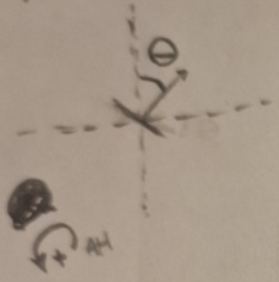
Estado do sistema:

$(x, y) \to$ pos do centro de massa

θ: ângulo de rotação (pitch) em rel ao eixo vertical.

$\dot{x}, \dot{y}, \dot{\theta} \to$ velocidades lineares e angulares

$\dot{\theta} \to \omega \left[\frac{rad}{s}\right]$

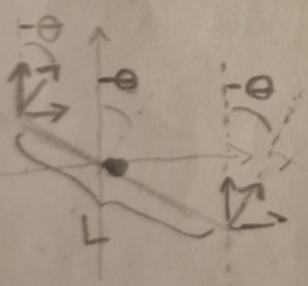


Entradas:

F_1 e F_2: força de cada motor

Obs: $\sin(-\theta) = -\sin\theta$ (ímpar)
$\cos(-\theta) = \cos\theta$ (par)

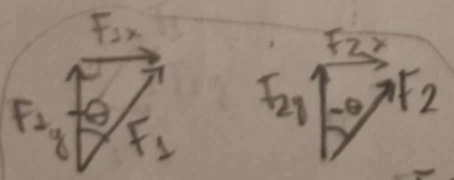
$$F_r = m \cdot a \to F_1 + F_2 - m \cdot g = m \cdot a$$



Em y:

$$m \cdot \ddot{y} = F_1 \cos\theta + F_2 \cos\theta - mg$$

$$\Rightarrow \boxed{\ddot{y} = \frac{(F_1 + F_2)}{m} \cdot \cos\theta - g}$$



$F_{1x} = -F_1 \sin\theta$ $\quad F_{2x} = -F_2 \sin\theta$

$F_{1y} = F_1 \cos\theta$ $\quad F_{2y} = F_2 \cos\theta$

Em x:

$$m \cdot \ddot{x} = -1(F_1 + F_2) \cdot \sin\theta$$

$$\Rightarrow \boxed{\ddot{x} = -1\left(\frac{F_1 + F_2}{m}\right) \sin\theta}$$

Cinem. Angular: $\tau = I \alpha \to \ddot{\theta}$

$$\Rightarrow \ddot{\theta} = \frac{\tau}{I} = (F_2 - F_1)\frac{L}{2} \cdot \frac{1}{I} = \frac{6(F_2 - F_1)}{m \cdot L}$$

$I = \frac{1}{12} m L^2$

∴ O sistema é descrito por

$$\begin{cases} \ddot{x} = -1 \cdot \dfrac{(F_1 + F_2)\sin\theta}{m} \\[2ex] \ddot{y} = \dfrac{(F_1 + F_2)\cos\theta}{m} - g \\[2ex] \ddot{\theta} = \dfrac{(F_2 - F_1)\cdot L}{2 \cdot I} \end{cases}$$

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vetor de estado z

$\dot{z} = f(z, u)$

↳ vetor de entrada

$u = [F_1 \quad F_2]$

$z = [x \quad y \quad \theta \quad \dot{x} \quad \dot{y} \quad \dot{\theta}]$

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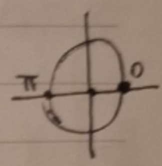
$$\dot{X} = F(X)$$

$$\frac{d}{dt}\begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ -\frac{\sin\theta}{m}(F_1+F_2) \\ \frac{\cos\theta}{m}(F_1+F_2) - g \\ \frac{(F_2-F_1)\cdot L}{2\cdot I} \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} \quad u = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

$$\dot{X} = 0 \Leftrightarrow \quad \dot{x} = 0, \quad \dot{y} = 0, \quad \dot{\theta} = 0$$



assumindo que $F_1 + F_2 \neq 0$

$$-\frac{\sin\theta}{m}(F_1+F_2) = 0 \Rightarrow -\sin\theta = 0 \Rightarrow \sin\theta = 0$$

$$\theta = 0, \pi, 2\pi, \dots = k\pi, \ k \in \mathbb{Z}$$

escolhendo $\boxed{\theta = 0}$

$$\frac{\cos\theta}{m}(F_1+F_2) - g = 0 \Rightarrow \boxed{(F_1+F_2) = m\cdot g}$$

$\theta = 0$

$$\frac{(F_2-F_1)L}{2I} = 0 \Rightarrow F_2 - F_1 = 0 \Rightarrow \boxed{F_2 = F_1} \rightarrow F_1 = \frac{mg}{2} = F_2$$

Logo, equilíbrio ocorre para:

$$X_{eq} = \begin{bmatrix} x \\ y \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad e \quad u_{eq} = \begin{bmatrix} \frac{mg}{2} \\ \frac{mg}{2} \end{bmatrix}$$

Calculando a Jacobiana para o sistema linearizado em torno do equilíbrio.

$$A = \frac{\partial F}{\partial X}\Big|_{eq} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{(F_1+F_2)}{m} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$-\frac{\cos(\theta_{eq})(F_1+F_2)}{m}$$

$$-\frac{\sin(\theta_{eq})(F_1+F_2)}{m}$$

$\theta = 0$

$= -g$

$$B = \left.\frac{\partial F}{\partial u}\right|_{eq} = \cancel{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\frac{\sin\theta_{eq}}{m} & \frac{\cos\theta_{eq}}{m} \\ -\frac{L}{2I} & \frac{L}{2I} \end{bmatrix}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\frac{\sin\theta}{m} & -\frac{\sin\theta}{m} \\ \frac{\cos\theta}{m} & \frac{\cos\theta}{m} \\ -L/2I & L/2I \end{bmatrix}$$

Para $\theta_{eq} = 0$:

Sistema Linearizado:

$\dot{X} = AX + Bu$

$$\underbrace{\frac{d}{dt}\begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}}_{\dot{X}} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -g & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}}_{X} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1/m & 1/m \\ -L/2I & L/2I \end{bmatrix}}_{B} \underbrace{\begin{bmatrix} F_1 \\ F_2 \end{bmatrix}}_{u}$$

Calculando autovalores de A:

$$\det(A - \lambda I) = 0 \iff \det\begin{pmatrix} -\lambda & 0 & 0 & 1 & 0 & 0 \\ 0 & -\lambda & 0 & 0 & 1 & 0 \\ 0 & 0 & -\lambda & 0 & 0 & 1 \\ 0 & 0 & -g & -\lambda & 0 & 0 \\ 0 & 0 & 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & 0 & 0 & -\lambda \end{pmatrix} = 0$$

1 π 180 180° π rad 1 rad

$$\iff -\lambda(-1)^2(-\lambda(-1)^2(-\lambda(-1)^8(-\lambda(-1)^6 \cdot \det\begin{pmatrix} -\lambda & 0 \\ -g & -\lambda \end{pmatrix} = 0.$$

$$\iff \lambda^4(\lambda^2) = 0 \Rightarrow \boxed{\lambda^6 = 0}$$

↳ todos os autovalores são 0.

Eq. Não hiperbólico → Não há λ c/ parte real $\neq 0$

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Da eq. linearizada ($\dot{X} = AX + Bu$):

$\ddot{x} = -g\cdot\theta_0$

$$\frac{d^2x}{dt^2} = -g\theta_0 \Rightarrow \frac{d\dot{x}}{dt} = -g\theta_0 \Rightarrow \int 1\,d\dot{x} = -g\theta_0\int 1\,dt$$

$$\Rightarrow \dot{x} = -g\theta_0 t + C \quad , \quad C \in \mathbb{R}$$

$$\Rightarrow \dot{x} = -g\theta_0 t + \dot{x}_0$$

$$\Rightarrow \frac{dx}{dt} = -g\theta_0 t + \dot{x}_0 \Rightarrow \int 1\,dx = -g\theta_0\int t\,dt + \dot{x}_0\int 1\,dt$$

$$\Rightarrow x(t) = -\frac{g\theta_0 t^2}{2} + \dot{x}_0\cdot t + C$$

$$\Rightarrow x(t) = -\frac{g\theta_0 t^2}{2} + \dot{x}_0\cdot t + x_0$$

Tbm da eq. linearizada:

$$\ddot{\theta}(t) = 0 \Rightarrow \dot{\theta}(t) = \dot{\theta}(0)$$

$$\Rightarrow \theta(t) = \dot{\theta}(0)\cdot t + \theta(0)$$

$$C = [B \ AB \ A^2B \ A^3B \ A^4B \ A^5B]$$

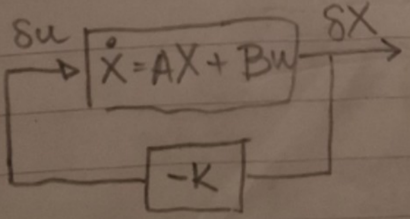
→ rank(C) = 6 → Controlável!!

K = place(A, B, [-1, -1, -1, -1, -1, -1])

$u = -K\cdot X$ como linearizamos em torno do equilíbrio:

→ $\delta u = -K\,\delta X \Rightarrow (u - u_{eq}) = -K(X - X_{eq})$

→ $u = u_{eq} - K(X - X_{eq})$



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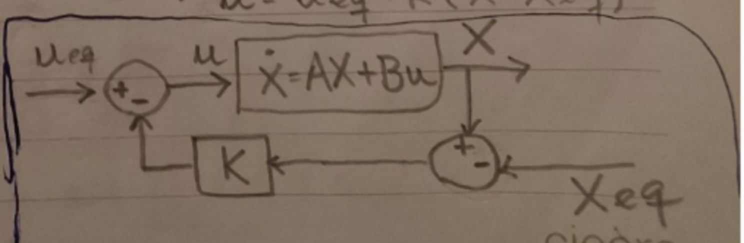

cicèro